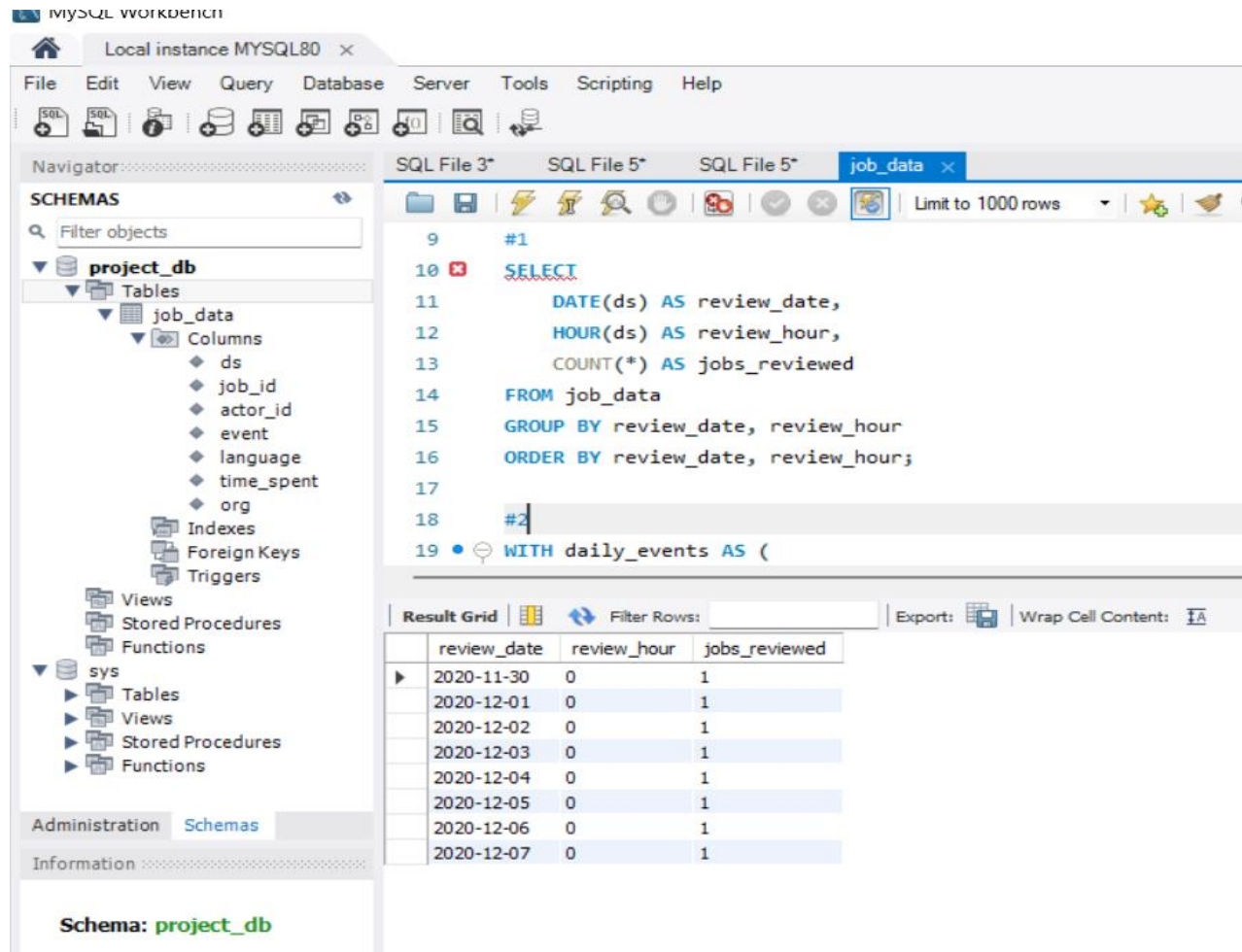


Case 1

- The dataset was raw where the ds column is masked and in text form .So to make it readable I changed into yyyy_mm_dd Form because mysql will understand in this form .
- Later i imported the csv file to mysql database.
- After that provided the insights for the given tasks through sql queries

A.Jobs Reviewed Over Time : Calculate the number of jobs reviewed per hour for each day in November 2020.



The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the 'project_db' database with its tables: 'job_data', 'sys', 'Tables', 'Views', 'Stored Procedures', and 'Functions'. The 'job_data' table is selected, showing its columns: 'ds', 'job_id', 'actor_id', 'event', 'language', 'time_spent', and 'org'. The main editor shows a SQL query in the 'job_data' tab:

```
9 #1
10 SELECT
11     DATE(ds) AS review_date,
12     HOUR(ds) AS review_hour,
13     COUNT(*) AS jobs_reviewed
14 FROM job_data
15 GROUP BY review_date, review_hour
16 ORDER BY review_date, review_hour;
17
18 #2
19 WITH daily_events AS (
```

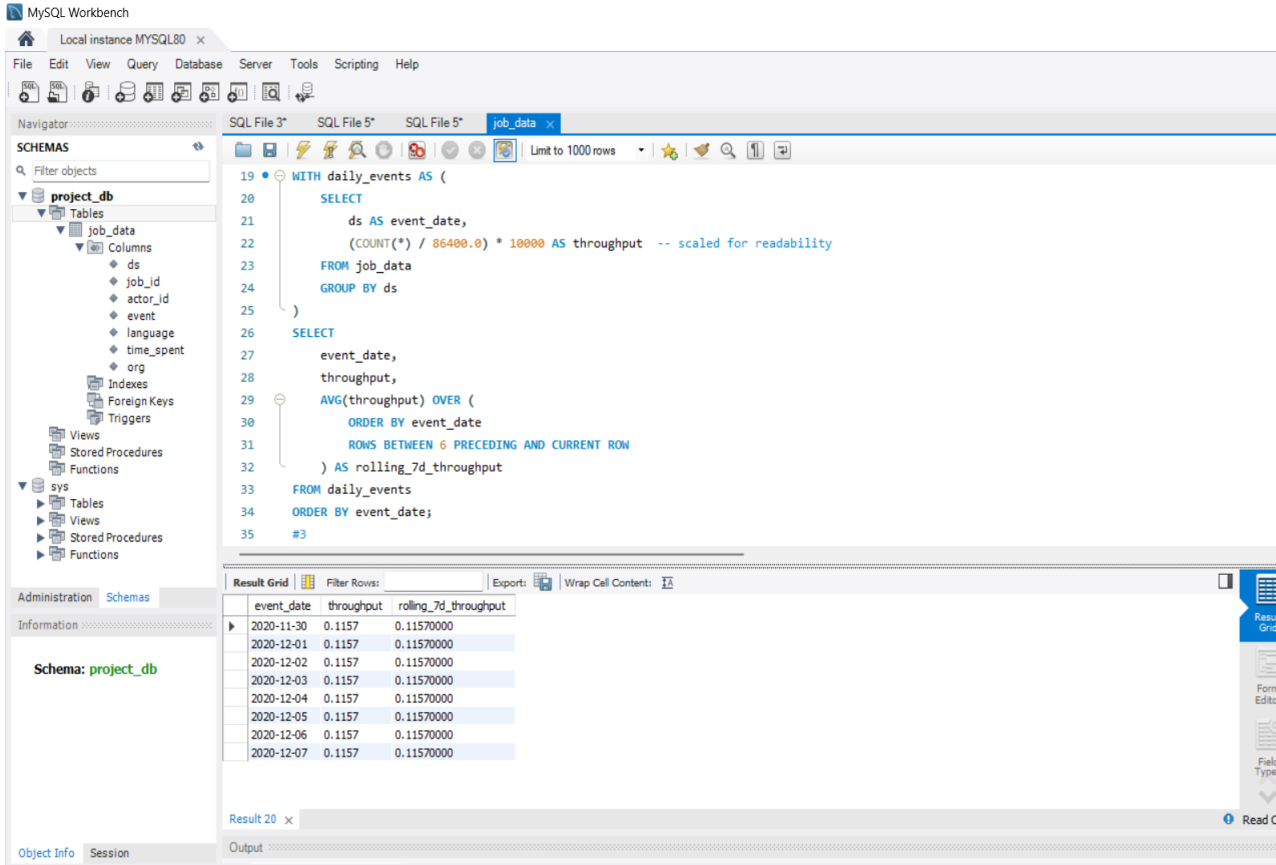
Below the query editor, the 'Result Grid' shows the results of the query:

review_date	review_hour	jobs_reviewed
2020-11-30	0	1
2020-12-01	0	1
2020-12-02	0	1
2020-12-03	0	1
2020-12-04	0	1
2020-12-05	0	1
2020-12-06	0	1
2020-12-07	0	1

The 'Schema: project_db' is displayed at the bottom left.

- As we didn't have the hour in the data set that's why it showing the review hour has zero
- **Insights:** the job reviewed per day is one

B.Throughput Analysis :Calculate the 7-day rolling average of throughput (number of events per second)



The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays the 'project_db' database with its tables: 'job_data', 'sys', and 'sys'. The 'job_data' table is selected, showing its columns: 'ds', 'job_id', 'actor_id', 'event', 'language', 'time_spent', 'org', 'Indexes', 'Foreign Keys', and 'Triggers'. The main editor displays a SQL query that calculates a 7-day rolling average of throughput. The query is as follows:

```
19 WITH daily_events AS (  
20     SELECT  
21         ds AS event_date,  
22         (COUNT(*) / 86400.0) * 10000 AS throughput -- scaled for readability  
23     FROM job_data  
24     GROUP BY ds  
25 )  
26 SELECT  
27     event_date,  
28     throughput,  
29     AVG(throughput) OVER (  
30         ORDER BY event_date  
31         ROWS BETWEEN 6 PRECEDING AND CURRENT ROW  
32     ) AS rolling_7d_throughput  
33 FROM daily_events  
34 ORDER BY event_date;  
35 #3
```

The 'Result Grid' at the bottom shows the following data:

event_date	throughput	rolling_7d_throughput
2020-11-30	0.1157	0.11570000
2020-12-01	0.1157	0.11570000
2020-12-02	0.1157	0.11570000
2020-12-03	0.1157	0.11570000
2020-12-04	0.1157	0.11570000
2020-12-05	0.1157	0.11570000
2020-12-06	0.1157	0.11570000
2020-12-07	0.1157	0.11570000

Insights

- Consistent Throughput** – The throughput stays the same every day because only one job is reviewed daily.

- Rolling Average Matches Daily Values** – Since the job count doesn't change, the 7-day rolling average is the same as the daily throughput.

c. Language Share Analysis: Calculate the percentage share of each language in the last 30 days.

MySQL Workbench

Local instance MYSQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

project_db

Tables

job_data

Columns

ds

job_id

actor_id

event

language

time_spent

org

Indexes

Foreign Keys

Triggers

Views

Stored Procedures

Functions

sys

Tables

Views

Stored Procedures

Functions

Administration Schemas

Information

Schema: project_db

SQL File 3* SQL File 5* SQL File 5* job_data x

Limit to 1000 rows

```
33 FROM daily_events
34 ORDER BY event_date;
35 #3
36 WITH last_30_days AS (
37     SELECT *
38     FROM job_data
39     WHERE ds >= DATE_SUB(MAX(ds), INTERVAL 30 DAY)
40 )
41 SELECT
42     language,
43     ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM job_data), 2) AS pct_share
44 FROM job_data
45 GROUP BY language
46 ORDER BY pct_share DESC;
47 #4
48 SELECT
49     ds, job_id, actor_id, event, language, time_spent, org,
```

Result Grid

Filter Rows:

Export:

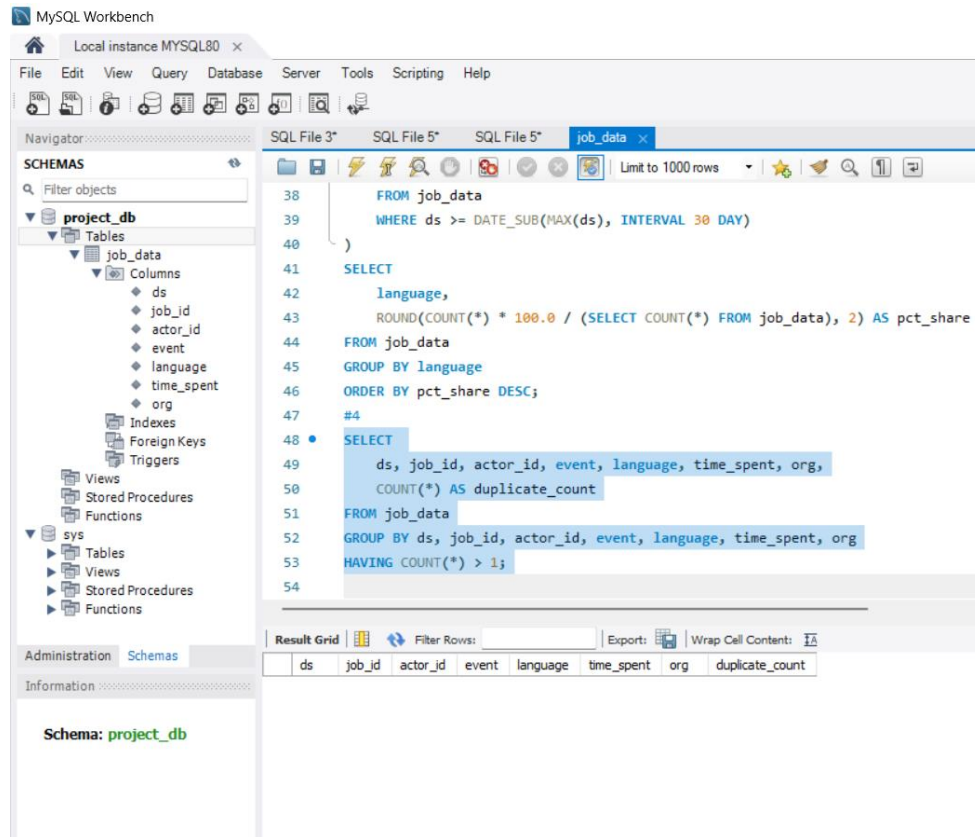
Wrap Cell Content:

language	pct_share
Persian	37.50
English	12.50
Arabic	12.50
Hindi	12.50
French	12.50
Italian	12.50

Insights

- Persian has the highest share in the dataset, making up 37.5% of the total jobs.
- Other languages like English, Arabic, Hindi, French, and Italian each contribute equally (12.5%).

C. Duplicate Rows Detection : Identify duplicate rows in the data.



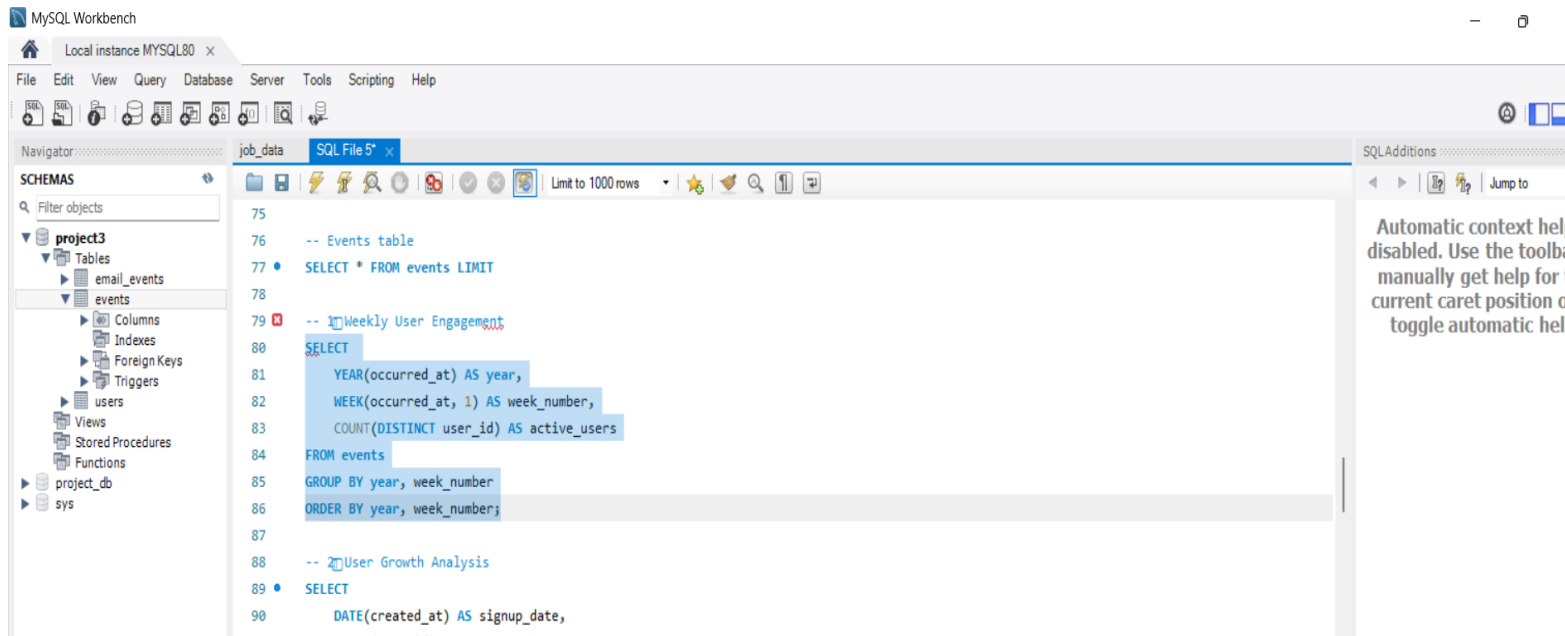
Insights :

- No duplicate rows in the job_data table

Case 2

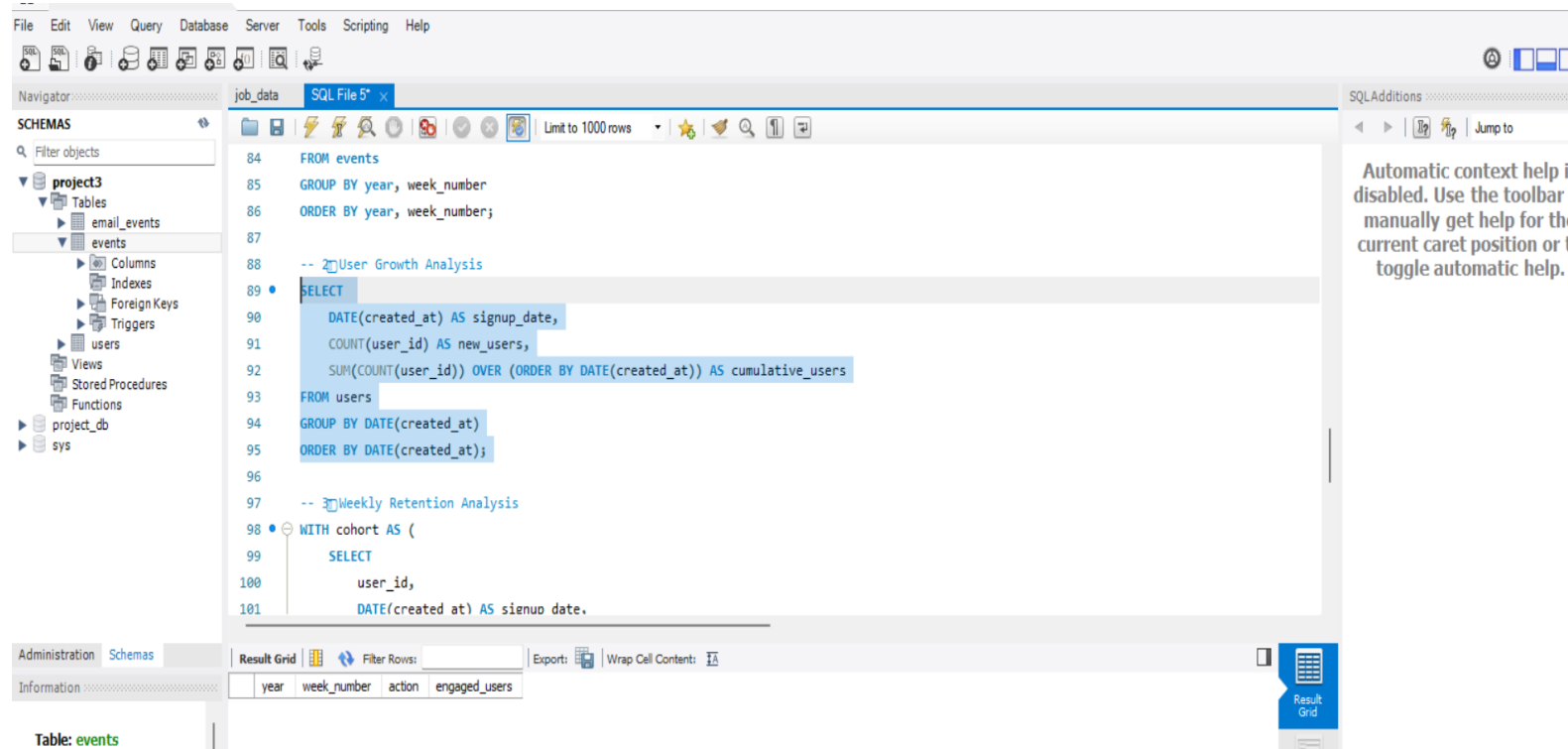
- The data sets have 3 tables in which the occurred_at, activated_at have the multiple dates formats.
- I converted them into text with formula `text(value,format)`
- I converted them into datetime in mysql using queries

A. Weekly User Engagement: Measure the activeness of users on a weekly basis.



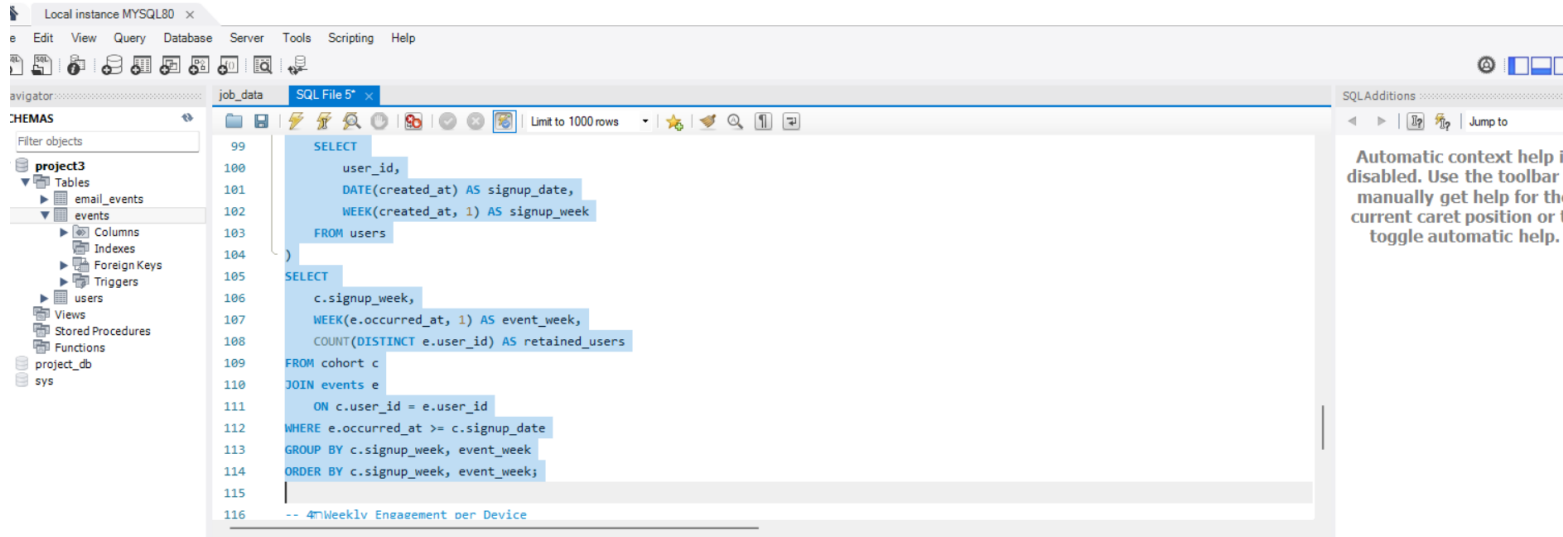
- **Insights:**
The number of active users varies week by week.
- Weeks with higher engagement indicate periods when users were more active on the platform, while drops may suggest issues, low interest, or external factors affecting usage.
- Monitoring these trends helps identify patterns and improve engagement strategies.

B. User Growth Analysis: Analyze the growth of users over time for a product.



- **Insight:** The daily new_users shows how many users joined each day, and cumulative_users shows the overall growth over time.
- Steady growth indicates successful acquisition, whereas sudden drops or spikes may point to marketing campaign effects or technical issues.

C. Weekly Retention Analysis: Analyze the retention of users on a weekly basis

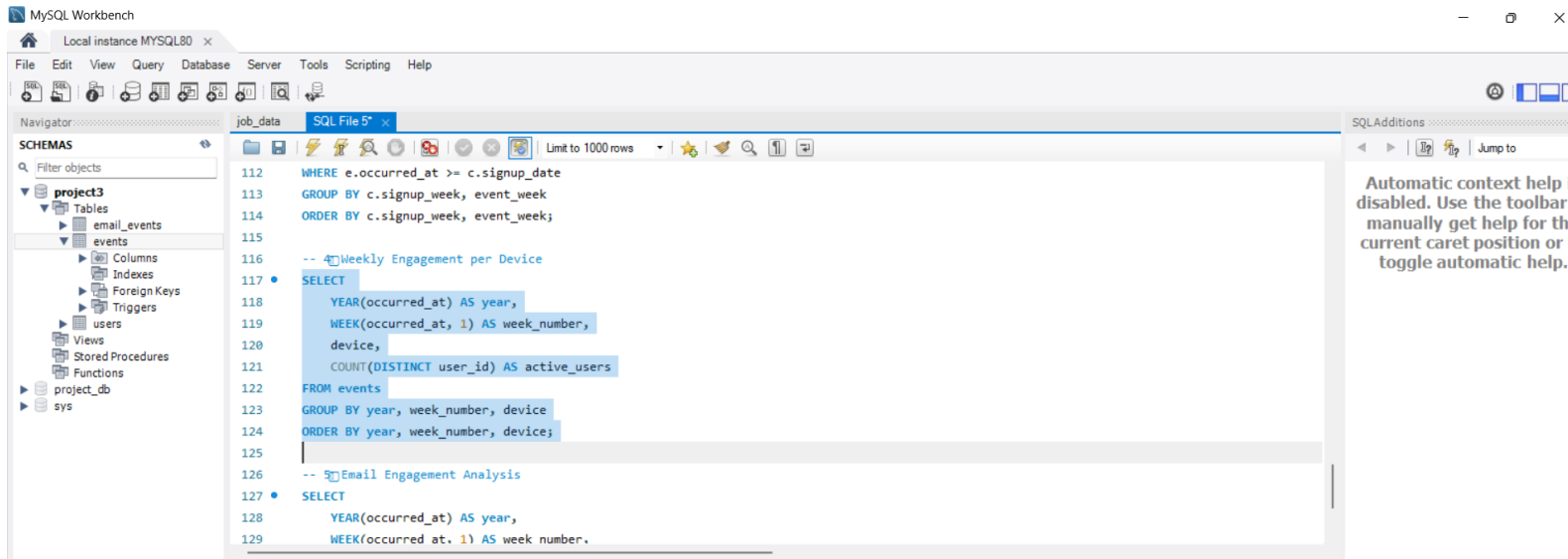


The screenshot shows a MySQL IDE interface with a file explorer on the left, a central SQL editor, and a right-hand pane. The file explorer shows a database named 'project3' with tables 'email_events' and 'events'. The SQL editor contains two queries. The first query (lines 99-104) selects 'user_id', 'signup_date', and 'signup_week' from the 'users' table. The second query (lines 105-116) is a more complex join between 'cohort' and 'events' tables, calculating 'event_week' and 'retained_users' (count of distinct user_ids) for each cohort and event week. The queries are grouped by cohort and event week, and ordered by cohort and event week. A comment at the bottom of the second query reads '-- 4 Weekly Engagement per Device'.

```
99      SELECT
100      user_id,
101      DATE(created_at) AS signup_date,
102      WEEK(created_at, 1) AS signup_week
103      FROM users
104  )
105  SELECT
106      c.signup_week,
107      WEEK(e.occurred_at, 1) AS event_week,
108      COUNT(DISTINCT e.user_id) AS retained_users
109  FROM cohort c
110  JOIN events e
111      ON c.user_id = e.user_id
112  WHERE e.occurred_at >= c.signup_date
113  GROUP BY c.signup_week, event_week
114  ORDER BY c.signup_week, event_week;
115
116  -- 4 Weekly Engagement per Device
```

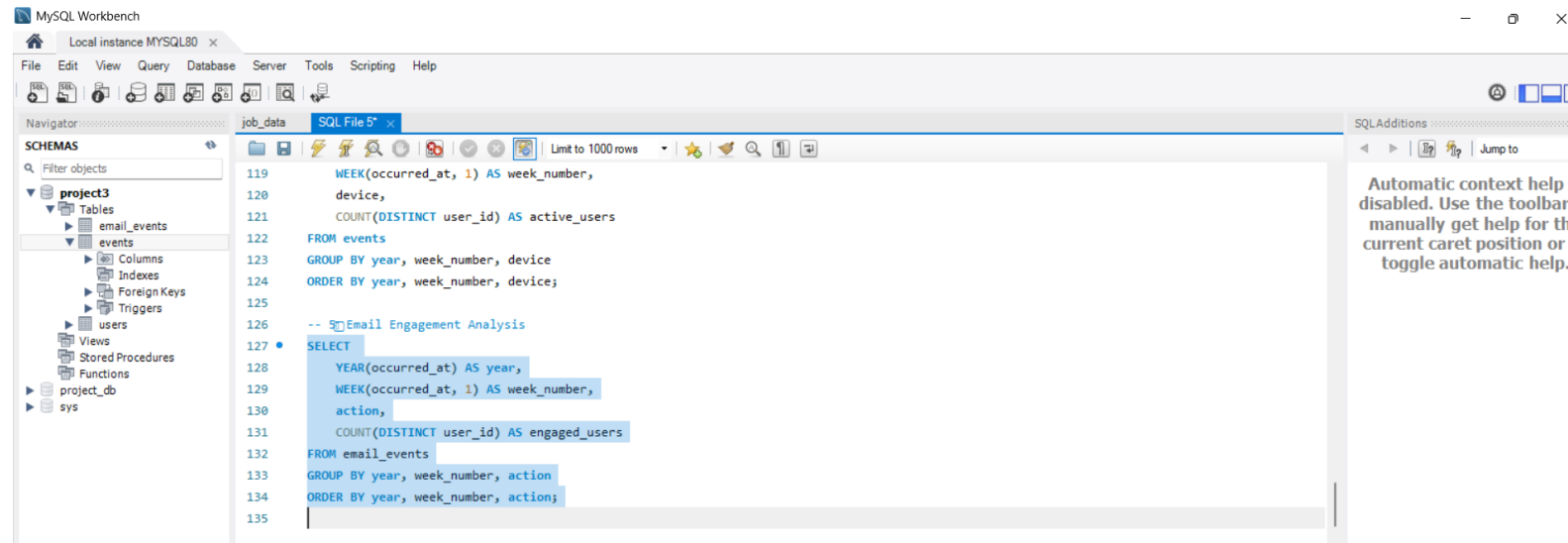
- **Insight:** Retention data reveals how well users continue using the product after signing up.
- High retention in the first few weeks indicates strong initial engagement, while declining retention suggests users may be losing interest.
- This helps prioritize features or campaigns to improve long-term engagement.

D. Weekly Engagement Per Device: Measure the activeness of users on a weekly basis per device.



- **Insight:** This shows which devices (mobile, desktop, tablet) users prefer each week.
- If one device consistently dominates, it guides design and feature prioritization.
- Unexpected drops or spikes for a device may indicate technical issues or changes in user behavior.

D. Email Engagement Analysis: Analyze how users are engaging with the email service.



- **Insight:** Tracking email interactions (action) weekly helps evaluate the effectiveness of email campaigns.
- A high number of opens or clicks indicates good engagement, whereas low engagement may suggest the need for better content, timing, or targeting strategies.

Tech stack :

- Used the mysql workbench for as software and excel for the cleaning .
- Mysql I faced the problems in importing any how Imported it.
- So for that from csv I converted into xslv and made changes and converted into csv